1 Introduction

Recall a *knot* is the image K of a smooth embedding $S^1 \hookrightarrow S^3$. A knot K is *smoothly slice* when there is a smoothly embedded disk D^2 in the 4-ball D^4 whose boundary is K. Similarly, a knot K is *topologically slice* when it is the boundary of a locally-flat embedded disk in D^4. Any smoothly slice knot is topologically slice, but the converse need not be the case. There are 352.2 million prime knots in the 3-sphere with at most 19 crossings [Bur]. We study which of these knots are slice, in both the smooth and topological categories. While no algorithm is known for deciding whether a given knot is slice in either setting, we are able to determine it smoothly for all but $\approx$ 11,400 knots (0.003% or about 1 in 30,000) and topologically for all but $\approx$ 1400 knots (0.0004% or about 1 in 250,000), using a wide variety of tools and techniques. In particular, we show that some $\approx$ 1.6 million of these knots (0.46%) are smoothly slice and that $\approx$ 350.5 million are not even topologically slice (99.54%). More precisely:

1.1 Theorem. *Let $\mathcal{K}_{19}$ denote the nontrivial prime knots in the 3-sphere with at most 19 crossings, up to homeomorphism of S^3, so that $\#\mathcal{K}_{19} = 352{,}152{,}252$. Then:*

(1) For $S = 1{,}633{,}786$ (0.46%), the number of smoothly slice knots in $\mathcal{K}_{19}$ is in $[S, S + 11{,}383]$.

(2) For $T = S + 22{,}412 = 1{,}656{,}198$ (0.47%), the number of topologically slice knots in $\mathcal{K}_{19}$ is in $[T, T + 1429]$. There are at least 12,162 knots in $\mathcal{K}_{19}$ which are topologically slice but not smoothly slice.

Consequently, for $R = 350{,}494{,}625$ (99.53%), the number of knots in $\mathcal{K}_{19}$ that are not topologically slice is in $[R, R + 1429]$.

1.2 Ribbon knots. A *ribbon disk* for a knot K is a smooth slice disk D with image in $D^4 \setminus \{0\} \cong S^3 \times (0,1]$ so that the $(0,1]$-coordinate is a Morse function on D with no local maxes in the interior of D. A knot is *ribbon* when it has a ribbon disk. (Equivalently, a knot is ribbon when it is the boundary of a restricted kind of immersed disk $D \looparrowright S^3$ as in e.g. [Fox] or [Kaw, Definition 13.1.9].) As every ribbon knot is smoothly slice, one is led to ask:

1.3 Question [Fox]. *Are all smoothly slice knots ribbon?*

The slice-ribbon conjecture is that the answer to this question is yes. Here, all the smoothly slice knots identified in Theorem 1.1 are indeed ribbon.

1.4 Prior work. For knots with at most 12 crossings, the complete picture was finished only in 2020:

1.5 Theorem [LM, Pic]. *Of the* 2977 *nontrivial prime knots with at most 12 crossings, exactly 158 are smoothly slice and 159 are topologically slice. All of the smoothly slice knots are moreover ribbon.*

Several groups have worked to identify ribbon and/or slice knots with more crossings, including [Sto, Appendix A] and [GHMR]. The largest general list we are aware of is [GHMR], which identifies 1705 ribbon knots from among the 59,937 nontrivial prime knots with at most 14 crossings. We independently found the same 1705 knots as part of Theorem 1.1; see Section 2.9 for more.

For alternating knots, Owens and Swenton studied these questions at a vastly larger scale than Theorem 1.5 or [GHMR] using techniques specific to the alternating case:

1.6 Theorem [OS]. *Of the 1.2 billion nontrivial prime alternating knots with at most 21 crossings, at least* $R = 664{,}633$ *(0.05%) are ribbon and at most* $R + 3276$ *are smoothly slice.*

For alternating knots with at most 19 crossings, Owens and Swenton had only 86 knots whose smooth slice status was unknown. We resolved all of these in the course of proving Theorem 1.1 and moreover settled the question in the topological category:

1.7 Theorem. *Of the 51.3 million nontrivial prime alternating knots with at most 19 crossings, exactly* 82,043 *(0.16%) are ribbon and the rest are not topologically slice.*

1.8 Plausibly slice knots. We now outline the methods behind Theorem 1.1, starting with the basic topological obstructions from the 1960s. We call a knot *plausibly slice* when its Murasugi signature $\sigma(K) = 0$ and its Alexander polynomial $\Delta_K(t)$ satisfies the Fox-Milnor criteria, that is, there is an $f(t) \in \mathbb{Z}[t^{\pm 1}]$ with $\Delta_K(t) = f(t)f(t^{-1})$ up to a unit in $\mathbb{Z}[t^{\pm 1}]$. Any topologically slice knot is plausibly slice, see e.g. Theorems 8.18 and 8.19 of [Lic] or Corollary 12.3.13 of [Kaw]. There are 3,869,541 nontrivial plausibly slice prime knots with at most 19 crossings, which is 1.10% of the prime knots in that range; we use $\mathcal{PS}_{19}$ to denote these knots, and these are what we will study for the rest of this paper. (Algebraically slice knots are plausibly slice, and we suspect that nearly all knots in $\mathcal{PS}_{19}$ are actually algebraically slice. However, checking this is quite involved [Liv], and we are unaware of any software that handles the general case. Ad hoc methods applied to a random sample of 10,000 knots from $\mathcal{PS}_{19}$ not known to be topologically slice found that at least 99.9% are algebraically slice.)

Statistics of some basic properties of the knots in $\mathcal{PS}_{19}$ are shown in Figures 1, 2, and 3. Theorem 1.1 follows immediately from:

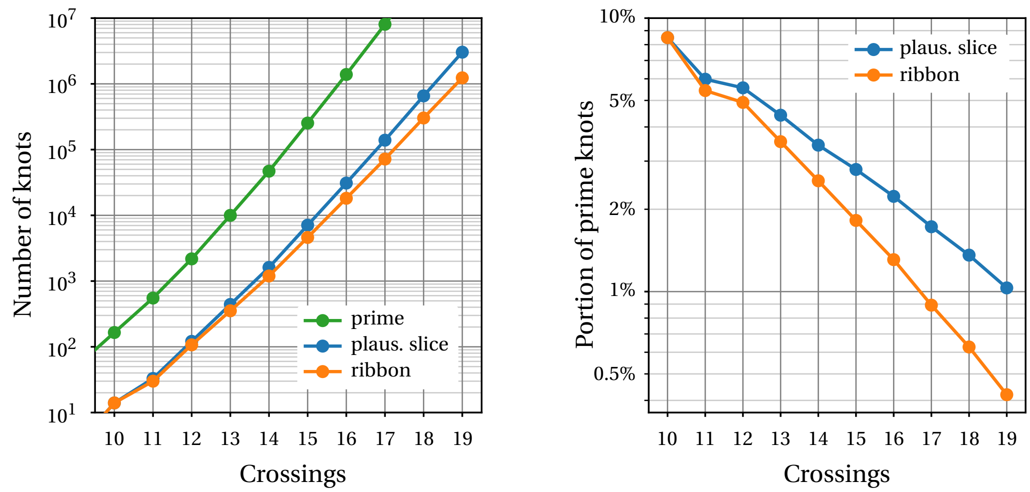


Figure 1. The number of prime plausibly slice and prime ribbon knots, as compared to all prime knots. For 20 crossings, we expect about 15 million prime plausibly slice knots and 5.5 million prime ribbon knots. (There are 1.8 billion prime knots with 20 crossings [Thi] and the plot at right suggests 0.8% are plausibly slice and 0.3% are ribbon.)

1.9 Theorem. *Of the knots in* $\mathcal{PS}_{19}$*, at least* 1,633,786 *are ribbon and at least* 2,224,372 *are not smoothly slice. Also, at least* 1,656,198 *are topologically slice and* 2,211,914 *are not topologically slice. This leaves* 11,383 *and* 1429 *unknowns in the smooth and topological settings respectively. There are at least* 12,162 *knots in* $\mathcal{PS}_{19}$ *that are topologically but not smoothly slice.*

Two methods handle all but 0.6% of the knots in Theorem 1.9:

(1) All 1,633,786 ribbon knots were identified using the process detailed in Section 2. Given a knot K, we did a combinatorial search for a sequence of band moves that turns K into the unlink and so specifies a ribbon disk. As needed, we looked for such moves in many different diagrams for K; these diagrams were generated by a more global method than just doing random Reidemeister moves. We also created a table of ≈ 12,100 two-component ribbon links which sometimes allowed us to certify a knot is ribbon after doing a single band move.

(2) The main slicing obstruction used was in the topological category, namely the technique of Herald, Kirk, and Livingston [HKL]. This method uses twisted Alexander polynomials to provide a computationally-efficient form of the Casson-Gordon invariants [CG2]. We introduce several refinements of this method in Sections 3.6, 3.9, 3.11, and 3.20. Collectively, such *HKL tests* obstruct 2,211,761 knots in $\mathcal{PS}_{19}$ from being topologically slice (Theorem 3.1).

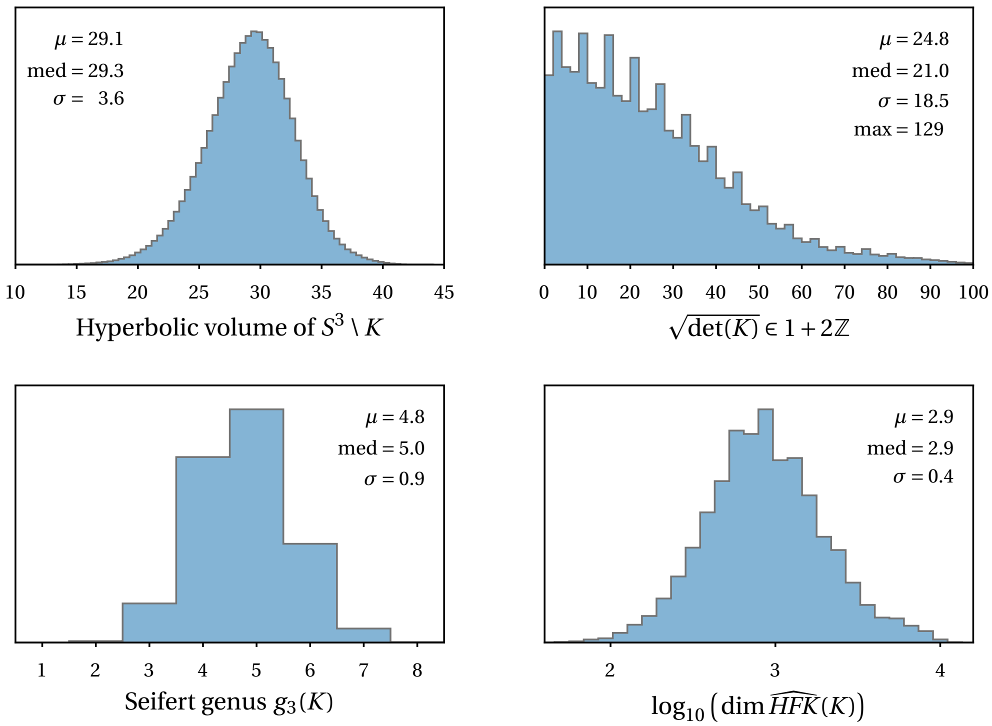


Figure 2. Some basic statistics about the 3.87 million plausibly slice knots. Here μ is the mean, med is the median, and σ is the standard deviation. The "spikes" in the plot of $\sqrt{\det(K)}$ are when that number is divisible by 3. The plot of the knot Floer homology $\widehat{HFK}$ is indistinguishable from the one of reduced Khovanov homology over $\mathbb{F}_2$ or $\mathbb{Q}$.

To handle the remaining 23,994 knots, we used the following methods:

(3) We computed many different smooth slice obstructions from Khovanov homology, knot Floer homology, and gauge theory; see Section 4 for details, but this includes the s-invariant in Khovanov homology and several of its generalizations. Collectively, these show that 333,912 knots (8.6% of $\mathcal{PS}_{19}$) are not smoothly slice (Theorem 4.1). However, only 10,631 of these knots (0.3% of $\mathcal{PS}_{19}$) are not already covered by the HKL tests in (2).

(4) Some 36 knots need special attention including the Conway knot and the 2-1 cable on the figure 8 knot, which are not smoothly slice by [Pic] and [DKMPS] respectively. Theorem 5.10 shows that 25 knots are not smoothly slice using the 0-friends method discussed below in Section 1.11. Theorems 6.2 and 6.3 each show a single knot is not smoothly slice. Finally, Theorem 6.6 shows nine alternating knots are not topologically slice.

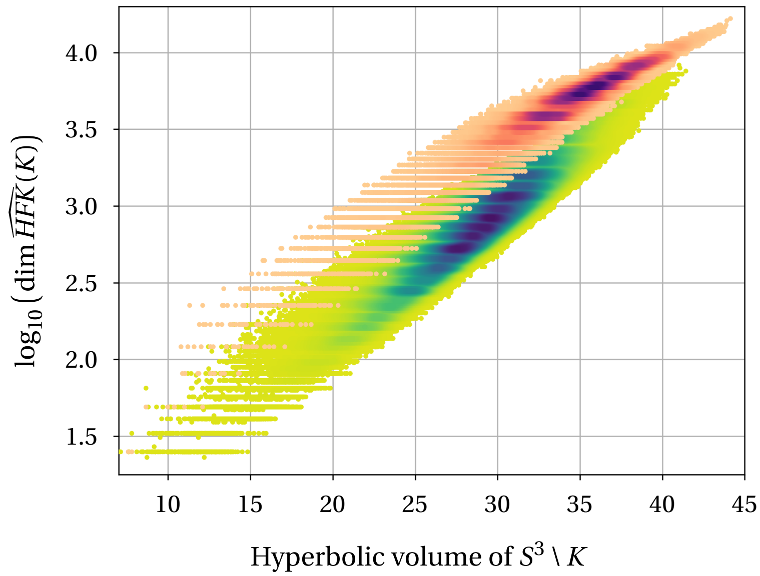


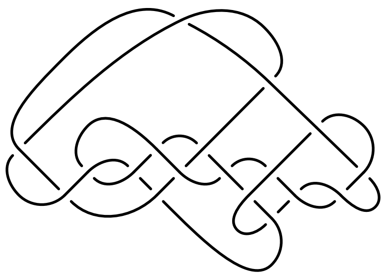
Figure 3. Comparing hyperbolic volume and the knot Floer homology $\widehat{HFK}$ for knots in $\mathcal{PS}_{19}$. Here, the alternating knots are the higher blob (peach-purple) and the non-alternating ones are the lower blob (green-blue); only 5.3% of the knots in $\mathcal{PS}_{19}$ are alternating, so this plot over-emphasizes them to show the relative behavior. This plot of $\widehat{HFK}$ is indistinguishable from the one for reduced Khovanov homology over $\mathbb{F}_2$ or $\mathbb{Q}$, so this matches the observation from [Kho, §8].

(5) The additional 22,412 topologically slice knots were primarily identified by having $\Delta_K = 1$, see [FQ, §11.7], or by the method of Section 2 producing a ribbon concordance to a knot with $\Delta_K = 1$. One knot, $18nh_{00000601}$, was shown to be topologically slice by the 0-friend method as part of Theorem 5.14.

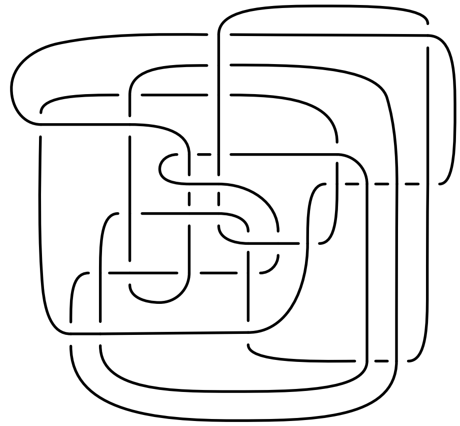
(6) The final ≈ 2000 knots we dealt with in Theorem 1.9 were handled in Section 7 by using the ribbon concordances between knots in $\mathcal{PS}_{19}$ produced in Section 2. For example, there are 1672 other knots in $\mathcal{PS}_{19}$ that are ribbon concordant to the Conway knot; since the Conway knot is not smoothly slice, the same is true for those 1672 knots.

1.10 Remark. It is worth noting that the topological HKL tests of (2) were about 6.6 times as effective as all the smooth obstructions in (3). Thus, in practice it seems that one should start with HKL tests rather the smooth invariants even if one only cares about slicing in the smooth category.

Also, the 10 knots in Theorems 6.2 and 6.6 as part of (4) are the only place where we rely directly on the vast literature of slice properties of particular knots. In all other cases, our computations are at least nominally independent of any prior work.



$K = 18nh_{00000601}$ in $\mathcal{PS}_{19}$



A 0-friend K' of K

Figure 4. A pair of 0-friends. The knot K has 18 crossings and the simplest diagram for K' known has 31 crossings. This pair will appear again in Section 5.13.

1.11 0-friends. The 0-surgery on a knot K in S^3 is the unique Dehn surgery where $b_1 > 0$. A pair of knots in S^3 are *0-friends* when their 0-surgeries are homeomorphic. A secondary focus of this paper, contained in Section 5, is to study 0-friends of knots in $\mathcal{PS}_{19}$. The motivation for this is twofold. First, and most tantalisingly, if K and K' are 0-friends where K is smoothly slice and K' is not, then there is an exotic smooth structure on S^4, disproving the smooth 4-dimensional Poincaré Conjecture; see [MP, §1]. Second, even with that conjecture unresolved, in favorable circumstances one can show that certain 0-friends must have the same smooth slice status; for example, this was used to show that the Conway knot is not smoothly slice [Pic].

There are only 101 pairs of knots in $\mathcal{PS}_{19}$ that are 0-friends (see Section 5.1), so we used the method of [DOR, §9.3] to generate more than 500,000 0-friends of the knots in $\mathcal{PS}_{19}$. Very roughly, starting with a knot K whose 0-surgery $Z_K = S^3_0(K)$ is hyperbolic, one drills out a closed geodesic in Z_K and tests whether the resulting manifold is the exterior of a knot in S^3. When it is, one can recover a diagram for the 0-friend using [DOR]. We outline in Section 5 why hyperbolic geometry suggests this method produces most small-to-medium 0-friends of knots in $\mathcal{PS}_{19}$; a sample pair of 0-friends found is shown in Figure 4.

Using this plethora of 0-friends and the methods of [Pic] and [MP], we show 25 knots in $\mathcal{PS}_{19}$ are not smoothly slice in Theorem 5.10. Following [MP], this involves producing an RBG link (see Definition 5.7 below) for each pair of 0-friends; we introduce a new computational method to search for such links in Section 5.11, again using [DOR].

We also found four pairs where we could determine the smooth sliceness of the

0-friend but not of the original knot in $\mathcal{PS}_{19}$, leading to:

1.12 Theorem. *If* $18nh_{00000601}$ *is not smoothly slice, or if any of* $\{16n68278, 17nh_{0010647}, 18nh_{00098198}\}$ *are smoothly slice, then there is an exotic smooth 4-sphere.*

All four knots in Theorem 1.12 are topologically slice, and $18nh_{00000601}$ is even smoothly slice in a homotopy 4-ball; see Theorem 5.14.

1.13 Ribbon concordances. Suppose K_0 and K_1 are knots in S^3. Recall from [Gor] that a *ribbon concordance from* K_1 *to* K_0 is a smooth annulus A embedded in $S^3 \times I$ with boundary components $K_0 \times \{0\}$ and $K_1 \times \{1\}$ so that the I-coordinate is a Morse function on A with no local maxima. In this case, we write $K_1 \geq K_0$. In particular, K is ribbon if and only if $K \geq$ unknot. Agol recently proved that $\geq$ gives a partial order on isotopy classes of knots [Agol]. Our search for ribbon disks produced 1,835,130 ribbon concordances $K_1 \geq K_0$; in 1,420,528 cases, K_0 was the unknot. In Section 7, we describe how these give evidence for conjectures that some invariants, such as the hyperbolic volume, are decreasing with respect to this partial order. A sampling of the ribbon concordances we found is shown in Figure 5.

1.14 Category alignment. We now turn to two situations where there seems to be no difference between topological and smooth sliceness. First, recall Lisca showed that a 2-bridge knot is ribbon if and only if it is smoothly slice [Lis]. Moreover, it is posited that a 2-bridge knot K is smoothly slice if and only if it is topologically slice [FM, Conjecture 1]. While there are 2-bridge knots whose topological and smooth slice genera differ [FM], our Theorem 1.7 above still compels us to propose:

1.15 Conjecture. *A prime alternating knot is smoothly slice if and only if it is topologically slice.*

Second, recall that a knot K is *fibered* when the exterior $E_K = S^3 \setminus \nu(K)$ fibers over the circle; here, the fiber surface is necessarily the unique minimum genus Seifert surface. Using knot Floer homology computed via [Sza], we found that 29.9% of the knots in $\mathcal{PS}_{19}$ are fibered. Per Table 5 below, we resolved both topological and smooth sliceness for all but 25 of the fibered knots in $\mathcal{PS}_{19}$. Thus, we were vastly more successful with fibered knots than nonfibered ones (factors of ≈ 200 and ≈ 25 in the smooth and topological settings respectively). This is despite not employing any tests specific to the fibered case, though some exist at least in theory [CG1, CL]. Regardless, in all but the 25 unknown cases, the topological and smooth sliceness matched for fibered knots, suggesting:

1.16 Conjecture. *A fibered hyperbolic knot is smoothly slice if and only if it is topologically slice.*

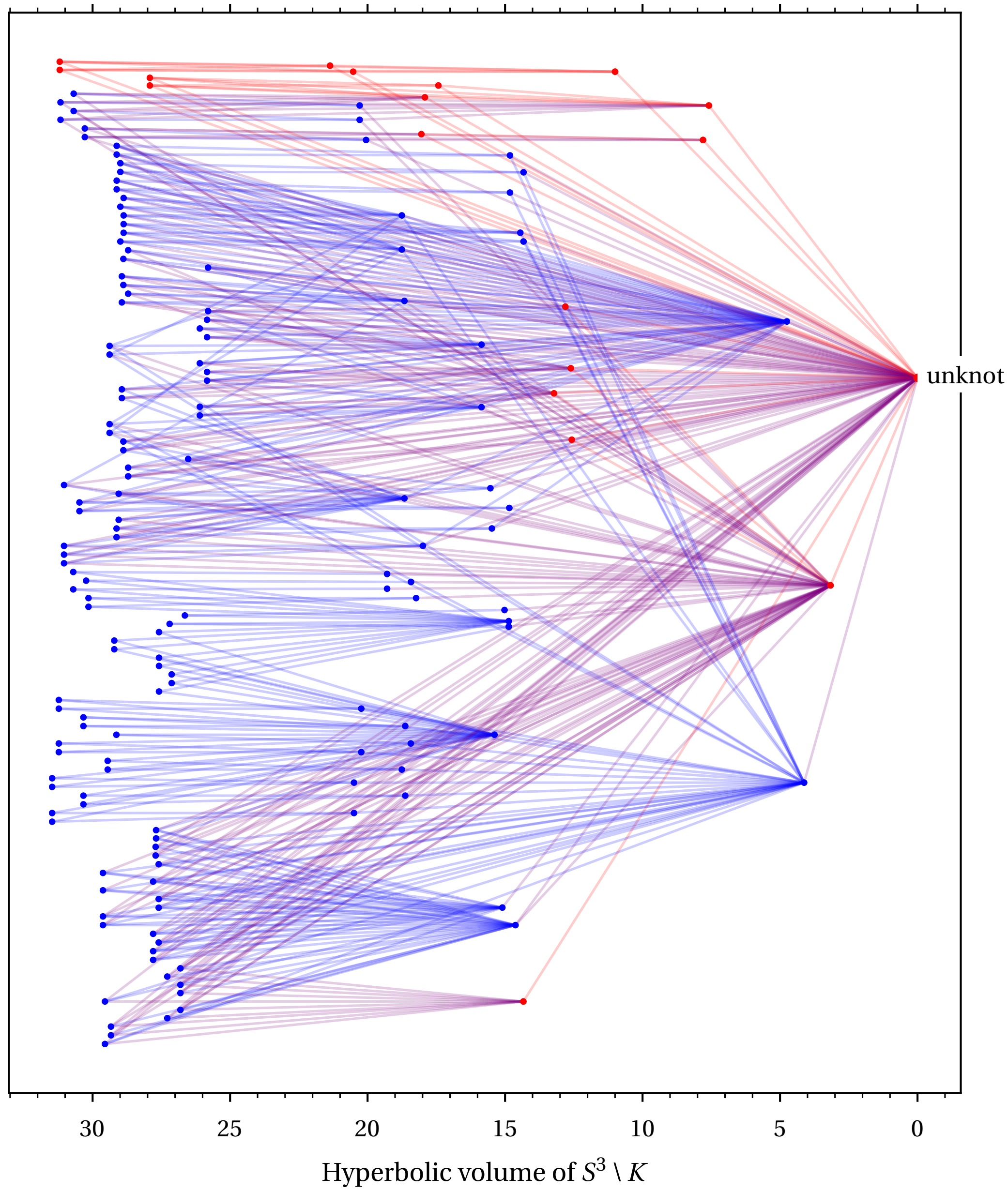


Figure 5. A visualization of ribbon concordances between 168 smoothly slice knots. Each knot is indicated by a dot, where alternating knots are red and other knots are blue. Each edge represents a ribbon concordance $K_1 \geq K_0$ where K_1 always corresponds to the dot farther to the left; that is, the hyperbolic volume of $S^3 \setminus K$ respects the ribbon partial order. The color of each edge is the average of its vertices. Note there are no $K_1 \geq K_0$ where K_1 is alternating and K_0 is not. See Section 7 and Figure 21 for more details.

Here, we restrict to hyperbolic knots because there is only a single non-hyperbolic fibered knot in $\mathcal{PS}_{19}$, namely $17ns_{29}$. Note that a fibered knot that is topologically but not smoothly slice would give rise to a counterexample either to the smooth 4D Poincaré conjecture or to the topological slice-ribbon conjecture discussed in the next subsection [Rub].

1.17 The slice-ribbon conjecture. To search for counterexamples to the slice-ribbon conjecture, one needs methods for showing that a knot K is smoothly slice without exhibiting a ribbon disk. Recall from e.g. [Tei, Lemma 2.5] that a knot K is smoothly slice if and only if there is a ribbon knot J so that $K \# J$ is ribbon. As recounted in Section 2.6, we tried this strategy for several small J. Taking a hint from how [HKL] found a slice disk for $K12a990$, we can sometimes choose J so that we know in advance that $K \# J$ is ribbon concordant to a knot that is simpler than K. Using this method, we discovered more than 500 smoothly slice knots where repeated passes of our ribbon disk search had come up empty. However, we eventually found ribbon disks for all of them; see Section 2.6 for details. Thus we have identified no smoothly slice knots that are not known to be ribbon.

However, the knot $K = 18nh_{00000601}$ from Theorem 1.12 is intriguing in the context of the slice-ribbon conjecture. Unless it and its 0-friend give a counterexample to the smooth 4D Poincaré conjecture, the knot K must be smoothly slice. However, we tried very hard to find a ribbon disk for K to no avail. This makes K a plausible candidate for a knot that is smoothly slice but not ribbon.

A version of the slice-ribbon conjecture in the topological category is as follows. Recall that a knot K is *topologically homotopy ribbon* when it bounds a locally flat disk $F \subset D^4$ so that the inclusion $S^3 \setminus K \hookrightarrow D^4 \setminus F$ is surjective on fundamental groups. (Any ribbon knot is topologically homotopy ribbon [Gor, Lemma 3.1].) The topological slice-ribbon conjecture is that every topologically slice knot is topologically homotopy ribbon. Of the topologically slice knots in Theorem 1.9, all but possibly $18nh_{00000601}$ are in fact topologically homotopy ribbon. This is because knots with $\Delta_K = 1$ are topologically homotopy ribbon by [FQ, Theorem 11.7B] and this is the basis for our primary method (5) above for showing that a knot is topologically slice.

1.18 Inscrutable knots. We next highlight some of the simpler knots not covered by Theorem 1.1 so that they can serve as challenges and benchmarks for future work in this area. Table 1 lists the knots of small crossing number whose smooth sliceness remains unknown; Table 2 is the topological analogue. Table 3 focuses on unknown knots with small hyperbolic volume and Table 4 lists the non-hyperbolic knots; together, these two tables includes all unknown knots in $\mathcal{PS}_{19}$ with Seifert genus 1. Finally, Table 5 lists all unknown knots in $\mathcal{PS}_{19}$ that are fibered.

$K13n65$	$K14n10011$	$K15n17501$	$K15n110439$	$K15n132965$
$K13n3871$	$K14n11063$	$K15n21905$	$K15n117232$	$K15n137921^*$
$K13n3872$	$K14n18909$	$K15n27582$	$K15n120055$	$K15n138033$
$K13n3897$	$K14n18911$	$K15n33471^*$	$K15n121598$	$K15n138051$
$K13n3936$	$K14n21673$	$K15n40402$	$K15n121834$	$K15n140327$
$K13n4582$	$K15n2810$	$K15n49735$	$K15n121916$	$K15n140449$
$K14n3713^*$	$K15n4646$	$K15n53931^*$	$K15n122603$	$K15n144034$
$K14n4425$	$K15n11287$	$K15n94339^*$	$K15n123414$	$K15n146982$
$K14n4621^*$	$K15n11568$	$K15n95989$	$K15n124496$	$K15n154389$
$K14n5486$	$K15n11570$	$K15n95995$	$K15n124640$	$K15n155464$
$K14n9023$	$K15n16056$	$K15n103703$	$K15n130504$	$K15n157903^*$

Table 1. Above are the 55 knots with at most 15 crossings where we could not determine whether they are smoothly slice. With the exception of the seven marked with a "$*$", these knots have $\Delta_K = 1$ and hence are topologically slice [FQ, §11.7]. We do not know whether any of the ones marked with "$*$" are topologically slice; they all have $\Delta_K = (t^2 - t + 1)^2$.

$K14n3713$	$K15n115646^*$	$16n254312$	$16n658612^*$	$16n889116$
$K14n4621$	$K15n137921$	$16n288832$	$16n679074$	$16n898244$
$K15n33471$	$K15n157903$	$16n312764^*$	$16n716954$	$16n911851$
$K15n53931$	$16n86850$	$16n357101$	$16n740455$	$16n914151$
$K15n73236^*$	$16n129260$	$16n374204$	$16n800378^*$	$16n977108$
$K15n77799^*$	$16n178564$	$16n429842$	$16n829217$	$16n982575$
$K15n94339$	$16n180683$	$16n478839^*$	$16n843878$	
$K15n99019^*$	$16n218932^*$	$16n481747$	$16n889114$	

Table 2. Above are the 38 knots with at most 16 crossings where we could not determine whether they are topologically slice. The 9 known not to be smoothly slice are marked with a "$*$". Some 30 of these knots have $\Delta_K = (t^2 - t + 1)^2$.

knot	vol	$g_3(K)$	sm	top	knot	vol	$g_3(K)$	sm	top
$19nh_{000000055}$	8.66	2			$19nh_{000003713}$	14.04	6		y
$19nh_{000000156}$	9.96	1			$19nh_{000003849}$	14.09	1		y
$17nh_{0000497}$	11.85	2			$19nh_{000004248}$	14.22	2		
$18nh_{00000597}$	11.92	3		y	$19nh_{000004269}$	14.23	2		y
$18nh_{00000601}$	11.93	5		n	$18nh_{00003640}$	14.36	3		y
$18nh_{00000707}$	12.20	1	n		$16n429842$	14.45	2		
$18nh_{00000752}$	12.30	2			$18nh_{00004014}$	14.49	2		y
$18nh_{00000957}$	12.63	2			$18nh_{00004122}$	14.53	2	n	
$K15n77799$	13.15	2	n		$19nh_{000005728}$	14.60	2	n	
$18nh_{00001632}$	13.31	1	n		$19nh_{000006042}$	14.67	2		
$18nh_{00001818}$	13.44	2			$17nh_{0004389}$	14.73	3		y
$16n481747$	13.59	2			$19nh_{000007003}$	14.85	3		y
$18nh_{00002591}$	13.90	2		y	$18nh_{00005355}$	14.87	3		y
$18nh_{00002656}$	13.93	2		y	$17nh_{0005136}$	14.94	3		y

Table 3. The hyperbolic knots in $\mathcal{PS}_{19}$ of smallest volume whose slice status is not completely known. Here, an "n" in the "sm" column means the knot is not smoothly slice, and a "y" or "n" in the "top" column indicates a known topologically slice status. All the unknown knots with $g_3(K) = 1$ appear either in this table or Table 4.

knot	structure	$g_3(K)$	smooth	top
$K15n115646$	Trefoil[3/2]	1	n	
$16n800378$	Trefoil[−1/4]	1	n	
$17ns_{29}$	Fig8[1]	2	n	
$18ns_{51}$	Trefoil[A(1/2, 4/3)]	2	n	
$18ns_{52}$	Trefoil[A(1/2, 3/5)]	2	n	
$18ns_{86}$	Fig8[−1/2]	1		y
$19ns_{006}$	Trefoil[11/6]	1	n	
$19ns_{018}$	Trefoil[15/4]	1	n	
$19ns_{101}$	Trefoil[A(−1/2, 5/7)]	2	n	
$19ns_{157}$	Trefoil[C(−1/2, 1/3, −1/2)]	2	n	
$19ns_{193}$	Trefoil[D(2, −3, −1/2)]	2	n	
$19ns_{244}$	Fig8[−1/3]	2		

Table 4. The 12 non-hyperbolic knots in $\mathcal{PS}_{19}$ whose slice status is not completely known. These are all satellite knots; see [Bur, §5.9] for how to read the structure column. Here an "n" in the "smooth" column means the knot is not smoothly slice, and a "y" in the "top" column means the knot is topologically slice. There are only 21 non-hyperbolic knots in $\mathcal{PS}_{19}$, so we had a much lower success rate in this setting. All the unknown knots with $g_3(K) = 1$ appear either in this table or Table 3.

knot	$g_3(K)$	smooth	top	knot	$g_3(K)$	smooth	top
$17nh_{4549325}$	6			$19nh_{000458099}$	7		
$17ns_{29}$	2	n		$19nh_{000616024}$	4		
$18nh_{00000601}$	5		n	$19nh_{000889975}$	4		
$18nh_{00068958}$	4			$19nh_{000941086}$	4		
$18nh_{00251375}$	4			$19nh_{003772023}$	4		
$18nh_{00446655}$	4			$19nh_{009668884}$	4		
$18nh_{00584034}$	4			$19nh_{010449461}$	4		
$18nh_{01814106}$	4			$19nh_{019331276}$	4		
$18nh_{01872227}$	4			$19nh_{019761670}$	4		
$18nh_{04892103}$	4			$19nh_{029859281}$	5		
$18nh_{10190577}$	4			$19nh_{032455745}$	4		
$19nh_{000054314}$	4			$19nh_{080376121}$	4		
$19nh_{000107587}$	4						

Table 5. The 25 *fibered* knots in $\mathcal{PS}_{19}$ whose slice status is not completely known. Here, the knot $17ns_{29}$ is not smoothly slice by Theorem 6.2, and $18nh_{00000601}$ is topologically slice by Theorem 5.14. The genus of the fibered Seifert surface is listed for each knot.

1.19 Code and data. Of necessity, Theorem 1.9 is the result of a large-scale computer computation. The main software tools we used are SnapPy [CDGW], KnotJob [Sch2], the HFK Calculator [Sza], and SageMath [Sage]. All data and code needed to check our results are permanently archived at [DG], and key parts have been incorporated into the development version of SnapPy. The latter will be part of SnapPy 3.3 but can be used now following the instructions in [DG]. We include in [DG] a quick-access list of PD codes for all 286 knots in $\mathcal{PS}_{19}$ mentioned in this paper by name.

Producing Theorem 1.9 consumed more than 100 CPU-years of computational time over the course of five calendar-years. However, it would take a tiny fraction of that to confirm our results as, for example, we saved a description of each ribbon disk found (see Section 2.10), and these can be quickly checked as valid. Similarly, the data includes the (m, p) parameters of each successful HKL test.

One method we used to try to ensure the correctness of the code is worth mentioning; for concreteness, we describe it in the context of slice obstructions. We initially ran each new slice obstruction on the whole of $\mathcal{PS}_{19}$, even those knots which had already been identified as ribbon. Recall $\mathcal{PS}_{19}$ is more than 42% ribbon knots, and we identified most of those early in our computations. If the code had a bug causing it to randomly report it had found this slice obstruction for 1 in 100,000 knots, we would have seen about 16 ribbon knots being flagged as “not smoothly slice”. Even near the end, when we were attacking the last 50,000 knots with very

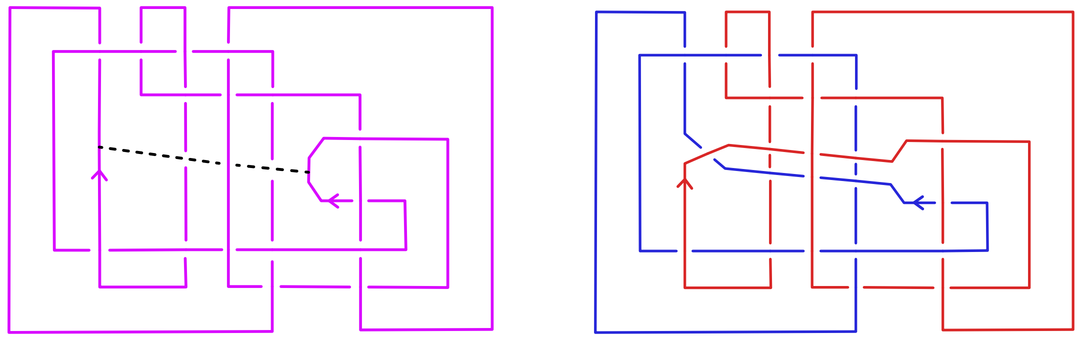


Figure 6. A band move on the knot $K = 16n424914$ at left resulting in the link at right. The band is added along the arc shown by the dotted line and has a half-twist so that the new link L has two components. In fact, L is the unlink, showing that K is ribbon.

time-consuming computations, each run included a substantial portion of knots known not to have the property being checked for as a trap for bugs.

1.20 Acknowledgements. We thank Robert Lipshitz, Ciprian Manolescu, Maggie Miller, Lisa Piccirillo, Jake Rasmussen, Danny Ruberman, Dirk Schütz, Zoltán Szabó, and Alex Zupan for helpful conversations and correspondence. This work was partially conducted at the Analytic and Geometric Aspects of Gauge Theory program held at SLMath (formerly MSRI). Dunfield was partially supported by US NSF grants DMS-1811156 and DMS-2303572 and by a fellowship from the Simons Foundation (673067, Dunfield). Gong was partially supported by NSF grants DMS-2055736 and DMS-2340465.

2 Searching for ribbon disks and concordances

This section is devoted to describing the process we used to show:

2.1 Theorem. *At least* 1,633,786 *knots in* $\mathcal{PS}_{19}$ *are ribbon and hence smoothly slice.*

First, we recall a standard description for ribbon disks that we use throughout. For a link L_0, a *band move* is a 1-handle attachment resulting in a link L_1 with one more component; see Figure 6. A band move is determined by an arc with endpoints in L_0 together with how the band twists around it. A band move encodes an embedded surface in $S^3 \times I$ whose interesting component is a pair of pants P with waist attached to L_0 and cuffs attached to L_1. Here, one arranges P to have a single critical point with respect to the I-coordinate, which has index 1. When a knot K has a sequence of k band moves resulting in an unlink U, it is ribbon: the cobordism in $S^3 \times I$ made up of k pants combines with disks capping off the components of U in D^4 to form a

5 Zero-friends

The 0-surgery on a knot K in S^3 is the unique Dehn surgery where $b_1 > 0$. A pair of knots in S^3 are *0-friends* when their 0-surgeries are homeomorphic. In this section, we study some 0-friends of the knots in $\mathcal{PS}_{19}$. The motivation for this is twofold. First, and most tantalisingly, if K and K' are 0-friends where K is smoothly slice and K' is not, then there is an exotic smooth structure on S^4, disproving the smooth 4-dimensional Poincaré Conjecture; see [MP, §1]. Second, even with that conjecture unresolved, in favorable circumstances one can show that certain 0-friends must have the same smooth slice status; for example, this was used to show that the Conway knot is not smoothly slice [Pic]. In Section 5.6, we use 0-friends to determine the smooth slice status of 25 knots in $\mathcal{PS}_{19}$. Section 5.13 gives 4 pairs of 0-friends where we can only determine smooth sliceness of one of the knots; these are thus candidates for constructing exotic smooth 4-spheres.

5.1 Finding 0-friends. There is an algorithm to determine when two 3-manifolds are homeomorphic (see e.g. [Kup]). While the known algorithms are unimplementable, in practice one can still usually decide this, especially in the presence of hyperbolic geometry [HW]. With 114 exceptions (0.003%), the knots in $\mathcal{PS}_{19}$ have hyperbolic 0-surgeries, so the first question is how many pairs of 0-friends there are in $\mathcal{PS}_{19}$ itself. The answer is just 101 pairs.[1] This is consistent with [KW], which found only 6 such pairs among all $\approx 313{,}000$ prime knots with ≤ 15 crossings. Note also that the 0-friend used in [Pic] is much more complicated than the Conway knot itself. Therefore, to usefully study 0-friends of knots in $\mathcal{PS}_{19}$, we need a technique that can generate them from a single given knot K.

We used the following method from [DOR, §9.3] to generate more than 500,000 0-friends of the knots in $\mathcal{PS}_{19}$; we explain in Section 5.4 why this should generate most of the interesting small-to-medium 0-friends. One example pair of 0-friends is shown in Figure 4.

(1) For each knot K whose 0-surgery $Z_K = S^3_0(K)$ is hyperbolic, we looked at all closed geodesics γ of length at most 3 that generated $H_1(Z_K;\mathbb{Z}) \cong \mathbb{Z}$, using [HW, CDGW]. (The reason we stopped at length 3 is explained in Section 5.4.)

(2) For each γ, we considered the exterior $X = Z_K \setminus \gamma$, which must itself be hyperbolic. The hyperbolic structure on X determines a finite list of Dehn fillings on

[1] The common 0-surgery is hyperbolic for 98 of these 101 pairs of 0-friends. There are only two nontrivial groups of mutual 0-friends, both with four knots: $\{K11n97,\ 16n86947,\ 18nh_{00000399},\ 18nh_{00000400}\}$ and $\{K11n67,\ 18nh_{00000254},\ 19nh_{000000631},\ 19nh_{000000632}\}$. See [DG] for all 186 knots involved.

it that might be S^3, and this can be used to identify whether X is the exterior of some unknown knot K' in S^3. (See e.g. [Dun] for more on this.)

(3) For each knot exterior X identified in Step 2, we used [DOR] to find an actual knot diagram for K'.

5.2 Remark. This method uses numerical computations about the hyperbolic structures that are not rigorous. However, each pair of 0-friends in our data must be genuine since filling, drilling, and finding knot diagrams are all combinatorial operations that are implemented at the level of manipulating triangulations of the relevant 3-manifolds. Such numerical issues instead mean that our list of 0-friends may not be complete according to the above description; still, we expect it contains 99% of such 0-friends.

5.3 Overview of 0-friends. We identified at least one 0-friend for 546,717 (14.1%) of the knots in $\mathcal{PS}_{19}$. (For these knots, we found an average of 1.5 0-friends, though we were not careful about avoiding duplicates.) The crossing number of the diagram of the 0-friend K' had mean 225.7 and median 154, with standard deviation σ of 219.2. (We make no claim that these are minimal crossing diagrams, though we suspect very few can be simplified to reduce the number of crossings by even 50%.) This makes most of these diagrams much too large to allow computing Khovanov homology and hence most of the invariants of Section 4. In contrast, the exteriors $E_{K'}$ are only modestly more complicated than the original E_K. The number of tetrahedra in the triangulation of $E_{K'}$ had mean and median 39.0 as compared to 31.0 for E_K, an increase of only 25% (here $\sigma = 7.91$ and $\sigma = 3.84$ respectively).

In the end, we were able to determine the smooth slice status for 78,507 0-friends of 56,319 knots in $\mathcal{PS}_{19}$; we denote these 0-friend knots by $\mathcal{ZF}$ and show information about their crossing number and hyperbolic volume in Figure 12. Some 82.0% of the knots in $\mathcal{ZF}$ are ribbon and 18.0% are not smoothly slice. (For all the knots in $\mathcal{PS}_{19}$ where we found a 0-friend, we had 71.5% ribbon and 28% not smoothly slice, with 0.4% unknown; this contrasts with $\mathcal{PS}_{19}$ as a whole where 42.2% are ribbon, 57.5% are not smoothly lice, and 0.3% unknown.) Initially, based on the computations described in Sections 2–4, there were 36 knots in $\mathcal{PS}_{19}$ of unknown smooth slice status where this was determined for a 0-friend. For 28 of these knots in $\mathcal{PS}_{19}$, we can use the 0-friend to determine the smooth slice status of the original; see Theorem 5.10 for more and Section 5.13 for the 8 mysterious remaining knots.

5.4 Hyperbolic biases. Our approach in Section 5.1 may seem to be one of convenience: we restrict to when Z_K is hyperbolic and then using that hyperbolic structure to guide the search for the 0-friend.

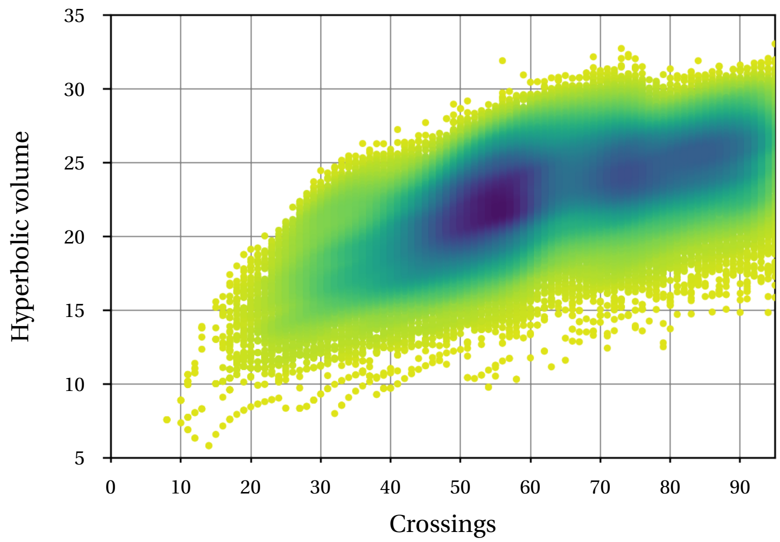


Figure 12. For each 0-friend K' in $\mathcal{ZF}$, this plot compares the crossing number of the simplest known diagram for K' with the volume of the hyperbolic structure on $E_{K'}$. The latter has mean 22.4, median 22.8, and $\sigma = 3.5$. This is about 9% more than the corresponding original knots in $\mathcal{PS}_{19}$; compare also Figure 2.

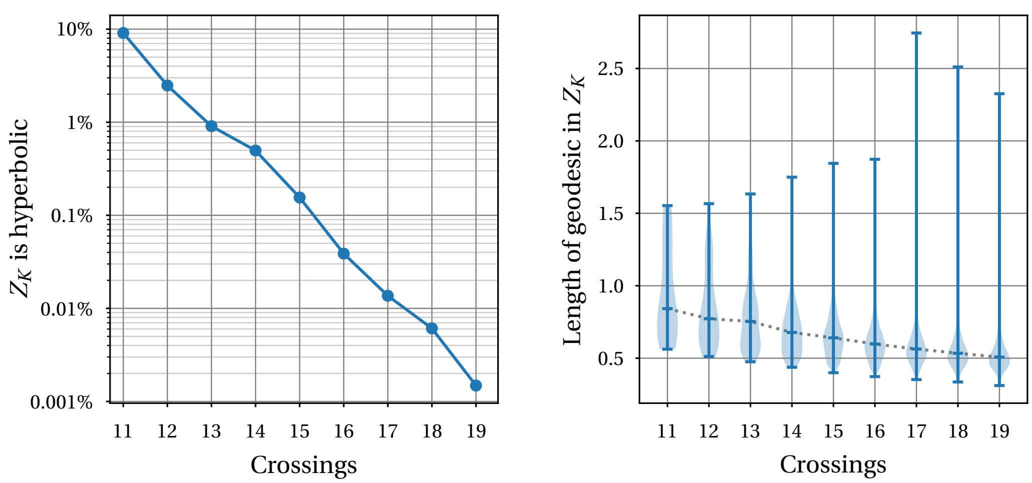


Figure 13. There are 93 knots K in $\mathcal{PS}_{19}$ whose exterior E_K is hyperbolic but Z_K is not hyperbolic. At left, we see the portion of such exceptional knots declines exponentially in the number of crossings, ending at 0.0015% (1 in 67,500) for 19-crossing knots. For $\mathcal{PS}_{19}$, every time Z_K was hyperbolic the core of the Dehn surgery solid torus was isotopic to a geodesic. The violin plot at right shows how the length of that geodesic changes as the number of crossings increases; the dotted line passes through the medians of each sample and the top and bottom lines are the maxes and mins.

We already noted that Z_K is hyperbolic for all but 114 knots K in $\mathcal{PS}_{19}$; this includes the 21 knots in $\mathcal{PS}_{19}$ where E_K itself is not hyperbolic. Moreover, the portion of such exceptional knots appears to decline exponentially in the number of crossings, as shown in the left plot of Figure 13. Thus insisting that Z_K be hyperbolic requires that we skip very few knots.

However, while any K' that is a 0-friend of K gives a corresponding knot in Z_K that generates $H_1(Z_K;\mathbb{Z})$, in Step 1 we restrict to those knots in Z_K that are *isotopic to geodesics* in its hyperbolic structure. Remarkably, whenever Z_K is hyperbolic for K in $\mathcal{PS}_{19}$, the core of the Dehn surgery solid torus in Z_K is indeed isotopic to a geodesic. We know of no geometric heuristic that suggests this pattern will not continue; if anything, we would expect it to become more marked if that were possible. Thus, when looking for a 0-friend, considering just the exteriors of geodesics in Z_K seems to be a very mild restriction, or possibly not one at all!

It remains to explain why we chose to limit our search to geodesics in Z_K of length at most 3. For the whole of $\mathcal{PS}_{19}$, the length of the core geodesic in Z_K had mean 0.54

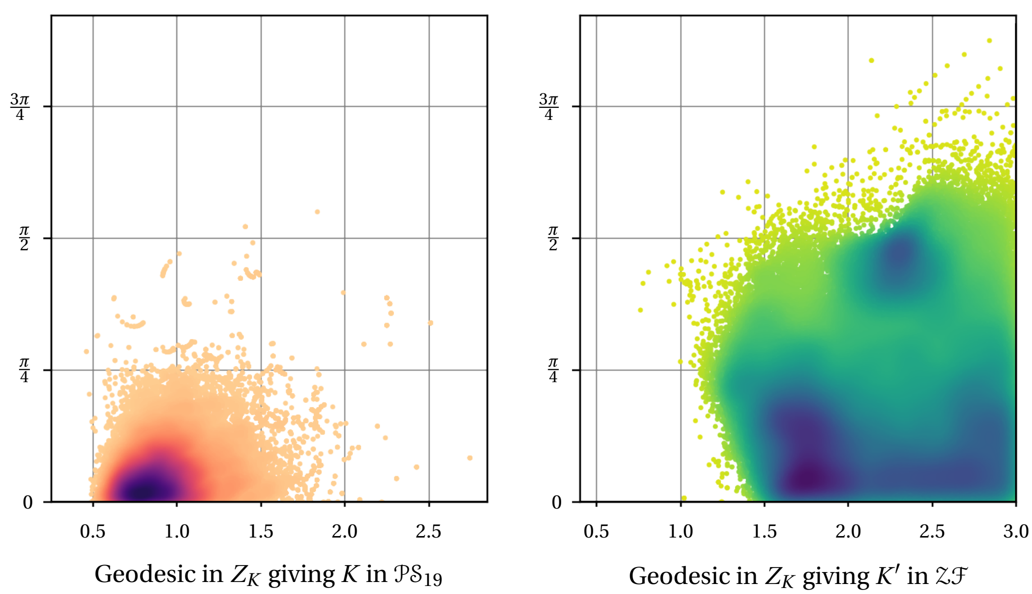


Figure 14. A closed geodesic γ in a hyperbolic manifold M has a length $L \in \mathbb{R}_{>0}$ and also a twist $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ recording the holonomy of going around γ; these are combined into its *complex length* $L + i\theta \in \mathbb{C}/(2\pi i\mathbb{Z})$. The above shows the complex lengths for the corresponding geodesics in Z_K for each pair of 0-friends K in $\mathcal{PS}_{19}$ and K' in $\mathcal{ZF}$. The complex length depends on the orientation of M but not of γ; as the original choice of K or $\overline{K}$ in [Bur] is somewhat arbitrary, we have normalized θ modulo π rather than 2π in both cases.

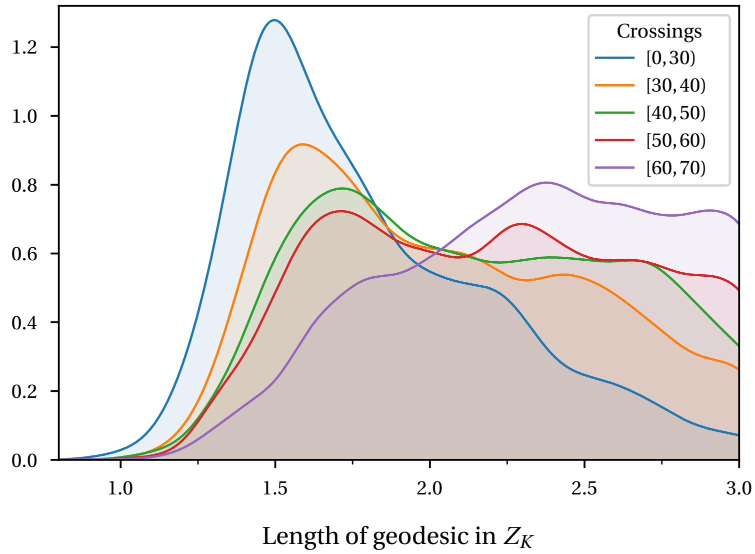


Figure 15. This plot explores how the length of the geodesic in Z_K that gives the 0-friend K' relates to the number of crossings of K'. Specifically, the 0-friends were grouped by crossing as indicated in the upper-right and then a smoothed histogram is shown for the lengths of the corresponding geodesics. The groups had between 2,659 and 25,316 knots each, where the size increases with the crossing number. Here, we include all 0-friends found for knots in $\mathcal{PS}_{19}$, not just those in $\mathcal{ZF}$.

and median 0.51, with $\sigma = 0.11$. No geodesic was longer than 2.75, and fewer than 0.01% were longer than 1.5. Moreover, as the number of crossings increased, the median length decreased as shown in the right plot of Figure 13, though a number of high outliers remain. Thus a cutoff of 3 is at least big enough to capture all of the 0-friends within $\mathcal{PS}_{19}$. On the other hand, the right plot in Figure 14 makes it clear there should be many 0-friends of knots in $\mathcal{PS}_{19}$ coming from geodesics in Z_K of length more than 3. The question is then how many "small-to-medium" 0-friends we are missing, where concretely we mean those with a a diagram of at most 50 crossings. Figure 15 suggests that we have found roughly 90% of those with less than 30 crossings, say 75% of those with less than 40 crossings, and perhaps 60% of those with less than 50 crossings. On the other hand, we likely have much less than 50% of those with crossing number in [50, 70).

It would be computationally feasible to raise the cutoff. As the number of geodesics grows exponentially in the length, this gets increasingly difficult, but considering those of length up to 4 (or perhaps 5) should be possible in most of these examples.

5.5 Hyperbolic aside. We next discuss an intriguing pattern in the hyperbolic geometry of these examples; it is not directly related to the main goal of using 0-friends to understand the knots in $\mathcal{PS}_{19}$, so you may want to skip ahead to Section 5.6. Our motivation here was to try to predict in advance which geodesics in Z_K give K' with a diagram with relatively few crossings, but we gained little purchase on this goal.

We start by considering the difference in volume between E_K and Z_K. By Thurston, one always has $\mathrm{Vol}(Z_K) < \mathrm{Vol}(E_K)$, and presuming we are in the situation that E_K is the complement of a closed geodesic γ in Z_K, the difference $\Delta\mathrm{Vol} = \mathrm{Vol}(E_K) - \mathrm{Vol}(Z_K)$ is typically controlled by the length L of γ, with $\Delta\mathrm{Vol} \approx \frac{\pi}{2}L$. The latter is always true asymptotically for sufficiently large Dehn fillings on E_K by [NZ]. Even for smaller Dehn fillings, as long as one can deform the hyperbolic structure on E_K to that of Z_K via hyperbolic cone manifolds, it is usually a good approximation because of Schäfli's formula; see e.g. [AST, §A.3]. Here, this pattern comes through very strongly in Figure 16. While for any knot K with a diagram with c crossings one has $\mathrm{Vol}(E_K) < v_{oct} \cdot c < 3.67c$, there is no converse to this and Figure 12 shows only a weak correlation between $\mathrm{Vol}(E_{K'})$, and the crossing number of the diagram for K'.

Less expected is the relationship between $\mathrm{Vol}(E_K)$ and L shown in Figure 17, namely that $\mathrm{Vol}(E_K)$ is roughly linear in $1/L$. Intriguingly, this holds only for the original K in $\mathcal{PS}_{19}$, not the 0-friends K' in $\mathcal{ZF}$. We now re-frame this phenomenon to remove reference to Z_K. Consider a horotorus T that cuts off a cusp neighborhood C of E_K; the Euclidean structure on T is unique up to dilation. Let μ and λ be the Euclidean translations corresponding to the meridian and longitude curves on T; one defines the *cusp shape* as $\tau_K = \lambda/\mu$, where by convention orientations are chosen so $\mathrm{Im}(\tau_K) > 0$. A simple calculation using Lemma 4.2 of [NZ] suggests that $L \approx \mathrm{Re}(2\pi i/\tau_K)$. (To see this, apply the lemma in the case $p = 0$ and $q = 1$, using that $v(u) = \tau_K u + O(u^3)$ by their equation (12), and $v(u) = 2\pi i$ by their equation (32).) This would imply $2\pi/L \approx |\tau_K|^2/\mathrm{Im}(\tau_K)$, suggesting that we compare $\mathrm{Vol}(E_K)$ with $|\tau_K|^2/\mathrm{Im}(\tau_K)$. Doing so results in the striking plot in Figure 18. This brings to mind [DJLTs], which relates $\mathrm{Re}(\tau_K)$ to σ_K, since plausibly slice knots have $\sigma_K = 0$, but we do not know a direct connection to Figure 18.

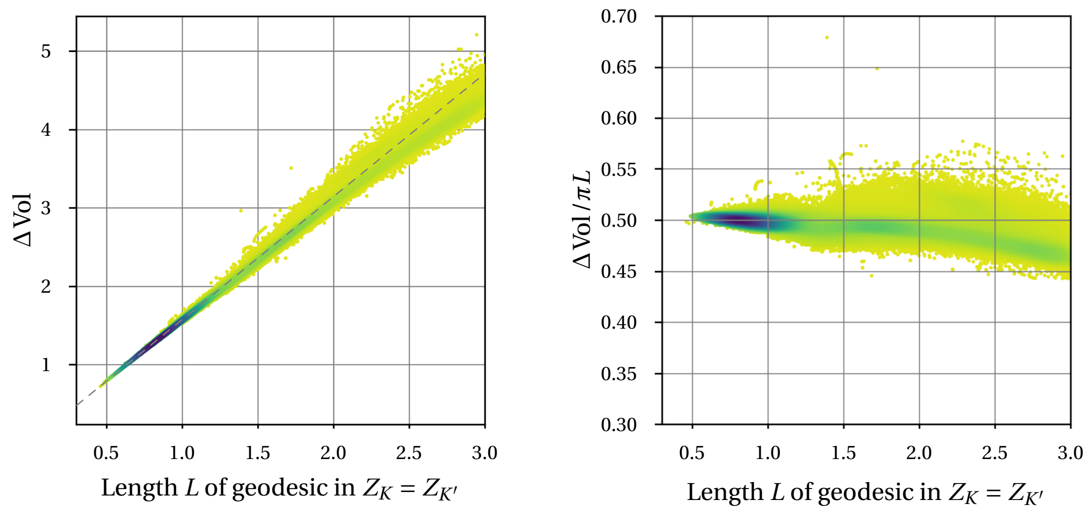


Figure 16. At left, for each K' in $\mathcal{ZF}$, we plot $\Delta\mathrm{Vol} = \mathrm{Vol}(E_{K'}) - \mathrm{Vol}(Z_{K'})$ against the length L of the corresponding geodesic in $Z_{K'}$ and we do the same for its corresponding 0-friend K in $\mathcal{PS}_{19}$. (From Figure 14, we see that typically $L > 1.5$ for K' and $L < 1.5$ for K. The dotted line is the prediction $\Delta\mathrm{Vol} = \frac{\pi}{2}L$. At right, the ratio $\Delta\mathrm{Vol}/\pi L$ is plotted instead; compare this with the figure in [AST, §A.3] which is for much smaller manifolds than those considered here.

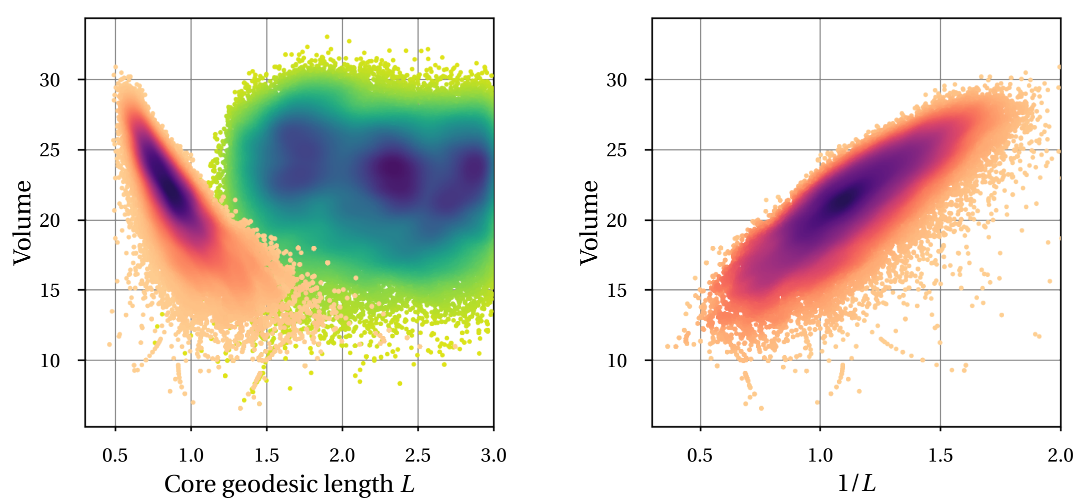


Figure 17. At left, for each pair of 0-friends K in $\mathcal{PS}_{19}$ and K' in $\mathcal{ZF}$, we plot the volume of E_K and E'_K against the length L of the corresponding geodesics in $Z_K = Z'_K$; the orange-red-purple blob where mostly $L \in [0.5, 1.5]$ corresponds to K in $\mathcal{PS}_{19}$ and the green-blue blob where mostly $L \in [1.5, 3]$ corresponds to K' in $\mathcal{ZF}$. Only the $\mathcal{PS}_{19}$ blob shows much structure, and that structure becomes even more evident at right where the x-axis is now $1/L$ rather than L.

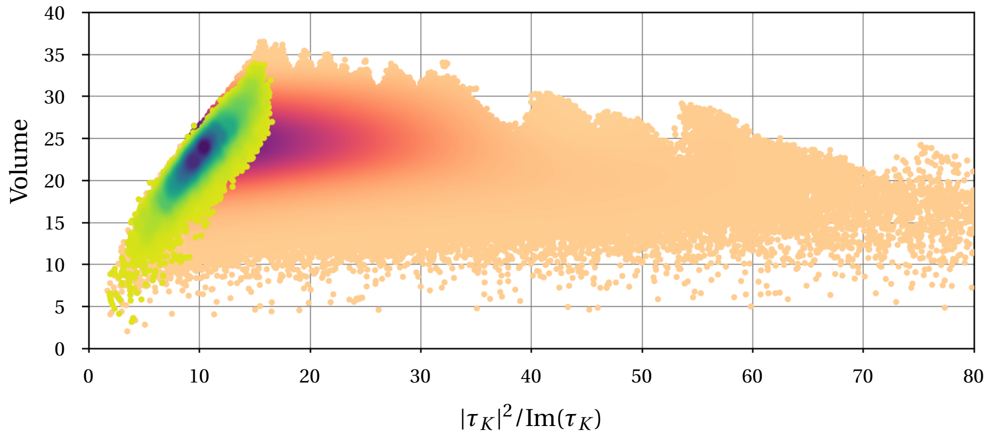


Figure 18. This plot compares the hyperbolic volume of E_K with the shown quantity based on the cusp shape τ_K. The smaller green-blue blob on the left corresponds to hyperbolic knots in $\mathcal{PS}_{19}$ with at most 16 crossings. The larger peach-purple blob corresponds to all hyperbolic knots with at most 16 crossings. The densest part of the larger blob is hidden behind the smaller one.

5.6 Applications. To explain what the 36 exceptional 0-friends say about the corresponding knots in $\mathcal{PS}_{19}$, we need to introduce the framework of red-blue-green (RBG) links, following [MP, §3].

5.7 Definition. An *RBG link* $L = R \cup B \cup G \subset S^3$ is a 3-component link, with rational framings r, b, g respectively, such that there exist homeomorphisms $\psi_B\colon S^3_{r,g}(R \cup G) \to S^3$ and $\psi_G\colon S^3_{r,b}(R \cup B) \to S^3$ and such that $H_1(S^3_{r,b,g}(L); \mathbb{Z}) \cong \mathbb{Z}$.

An RBG link describes a pair of 0-friends K_B and K_G where K_B is the image of B under ψ_B and K_G is the image of G under ψ_G; their common 0-surgery is $S^3_{r,b,g}(L)$. Conversely, any pair of 0-friends can be encoded by an RBG link [MP, Theorem 1.2]. An RBG link for the 0-friends in Figure 4 is shown in Figure 19.

We say that an RBG link is *super-special* when the two sublinks $R \cup B$ and $R \cup G$ are both Hopf links and the framings satisfy $b = g = 0$ and $r \in \mathbb{Z}$; this is a subset of the special RBG links of [MP]. The key facts for us are:

5.8 Theorem [MP]. *Suppose L is a super-special RBG link. If r is even, then K_B and K_G have homeomorphic traces; if $r = 0$, their traces are diffeomorphic. If instead r is odd and their common 0-surgery has trivial mapping class group, then their traces are not homeomorphic.*

Proof. This is in Sections 3 and 4 of [MP]; specifically, combine Theorems 3.7 and 3.13 with Lemmas 4.2 and 4.3, noting Remark 3.8. $\square$

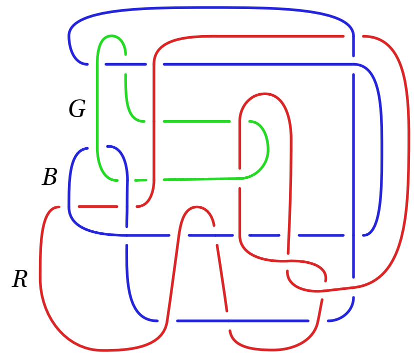


Figure 19. An RBG link L for the 0-friends K and K' in Figure 4. With $r = 1$, $b = g = 0$, one has $K' = K_B$ and $K = K_G$. The link L was found from (K, K') using the method of Section 5.11.

5.9 Theorem [Nak, Ren]. *Suppose L is a super-special RBG link. Suppose $s_{\mathbb{F}}(K_G) \neq 0$, where $s_{\mathbb{F}}$ is the Rasmussen s-invariant over a field $\mathbb{F}$. Then both K_B and K_G are not smoothly slice.*

Proof. As $s_{\mathbb{F}}(K_G) \neq 0$, of course K_G is not smoothly slice. For K_B, we use Theorem 3.13 of [Nak]; this result is conditional on Conjecture 2.15 in that paper, but this was established as Corollary 1.5 of [Ren]. If K_B was smoothly slice, since R is the unknot, Theorem 3.13 of [Nak] applies to say $s_{\mathbb{F}}(K_G) = 0$, a contradiction. □

The main result of this section is:

5.10 Theorem. *The 25 knots listed in Table 10 are not smoothly slice.*

The first two knots in Table 10 were already known not to be smoothly slice by [Pic] in the case of the Conway knot $K11n34$ and by unpublished work of Ciprian Manolescu, Lisa Piccirillo, and Julius Zhang in the case of $K13n866$ using a similar method to [Pic].

Proof of Theorem 5.10. For each of the 25 knots in the statment, the method of Section 5.11 below was used to find a super-special RBG link that encoded the relationship between the original K in $\mathcal{PS}_{19}$ and its 0-friend K'; this is recorded in Table 10.

For $K = 18nh_{00010270}$, as $r = 0$ in that RBG link we learn K and K' have diffeomorphic traces by Theorem 5.8. By the Trace Embedding Lemma (see e.g. [MP, Lemma 3.5]), it follows that K and K' have the same smooth slice status. Now K' is not smoothly slice, as $\boldsymbol{s}^{\mathrm{Sq}^1_o(K)}(K') \neq 0$, and so K cannot be either. The remaining 24 knots in Table 10 are covered by Theorem 5.9. □

knot	DT code of RBG link	r	K' notes
$K11n34$	vcdemhUSFVDCoKIMeJaRgtNpBql	0	$s_{\mathbb{F}_3} = 2$
$K13n866$	tcbiisTAbrNEomgQKhFiLdJpc	−1	$s_{\mathbb{F}_3} = 2$
$K15n25044$	xcdhlUkRJgNxsVpQMLFWhdBCoAITe	−4	$s_{\mathbb{F}_3} = 2$
$16n180537$	rcbdlGCARMoFBkQhiEPlDNJ	1	$s_{\mathbb{F}_3} = 2$
$16n74539$	tcbiiEnTOBDlRpqMfKgaCsjHi	−1	$s_{\mathbb{F}_3} = -2$
$17nh_{0001844}$	scbfkjFinLRAQObhDpeSKmGc	1	$s_{\mathbb{F}_3} = 2$
$17nh_{0002715}$	xcdoelHtPXVRDuSEaqFwCGmOibKnJ	0	$s_{\mathbb{F}_3} = -2$
$17nh_{0212094}$	sccdlrcesbjkOGlmifpqHnda	−1	$s_{\mathbb{F}_3} = 2$
$17nh_{0212095}$	qccdjenaQcmPOGldfjbkHI	−1	$s_{\mathbb{F}_3} = 2$
$18nh_{00010270}$	zcdkkgNKtZYuaQpCsrVDjXmwbOhlIFE	0	$\boldsymbol{s}^{\mathrm{Sq}^1_o(K)} \neq 0$
$18nh_{00166702}$	rcehemldoJRhqpEIFnabcgK	0	$s_{\mathbb{F}_3} = -2$
$18nh_{00610378}$	tccelDPAECTJSQGMNrKfHBlIO	1	$s_{\mathbb{F}_3} = 2$
$18nh_{00610381}$	ucecmKDGBAlRCOsEhTpqInFuMj	−1	$s_{\mathbb{F}_3} = 2$
$19nh_{000002588}$	ycblkuLFkRYOmWeSpAvNThIxCbGqDJ	−1	$s_{\mathbb{F}_3} = -2$
$19nh_{000003154}$	rcgfeClHoJPdBNqFaRMGkEi	3	$s_{\mathbb{F}_3} = -2$
$19nh_{000003570}$	uciffKlPtSBhoQUNrEdgCfIamJ	−5	$s_{\mathbb{F}_3} = 2$
$19nh_{000032808}$	xchigIfEOtUkwXmAsPjCVhQGdbNLR	0	$s_{\mathbb{F}_3} = -2$
$19nh_{000076489}$	wcbjkDftqRSMWovpGLKujcbEAhiN	−1	$s_{\mathbb{F}_3} = -2$
$19nh_{000018991}$	wcdijMJrFTLWiqgNUDaSvBhPkEoC	0	$s_{\mathbb{F}_3} = -2$
$19nh_{000066839}$	xcbnhMqXegIdTFOPcBrSWlavUHnJK	−1	$s_{\mathbb{F}_3} = -2$
$19nh_{000066841}$	ycbohEgYwUtsRxpMoJLVkiCAfbNdqh	1	$s_{\mathbb{F}_3} = -2$
$19nh_{000177115}$	zcdnhRqPoGwUVasCnTxyJzibDMEHflk	0	$s_{\mathbb{F}_3} = 2$
$19nh_{000177116}$	zcdscVuopGhWmqLZrFedcsiknXyTJaB	0	$s_{\mathbb{F}_3} = -2$
$19nh_{001336127}$	ycejjDCNAYSLrXtqWvPBOhkMuifGJE	−1	$s_{\mathbb{F}_3} = -2$
$19nh_{002457201}$	ycdkjOPHexuMLQyrcGVWASkIBNfTDj	0	$s_{\mathbb{F}_3} = -2$

Table 10. Some 0-friends that admit super-special RBG links. In each case, we can conclude that the original knot in $\mathcal{PS}_{19}$ is not smoothly slice.

5.11 Finding RBG links. Given 0-friends K and K' whose exteriors E_K and $E_{K'}$ are hyperbolic, we used the following method to search for a super-special RBG link L that expressed this fact. As in Section 5.1, we will use [DOR] to recover a diagram for L from its exterior as the very last step. In particular, we searched for a 3-cusped hyperbolic manifold Y with cusps labelled red, blue, and green with peripheral framings (μ_R, ν_R), (μ_B, ν_B), and (μ_G, ν_G) respectively so that:

(a) The Dehn filling $Y(\mu_R, \mu_B, \mu_G)$ is homeomorphic to S^3.

(b) We have $Y(\nu_R, \cdot, \nu_G) \cong E_K$ by a homeomorphism that takes (μ_B, ν_B) to the standard framing (μ_K, λ_K) for the knot exterior E_K. Likewise, $Y(\nu_R, \nu_B, \cdot) \cong E_{K'}$ where $(\mu_G, \nu_G) \mapsto (\mu_{K'}, \lambda_{K'})$. (Note this implies $Y(\nu_R, \nu_B, \nu_G)$ is the common 0-surgery of K and K'.)

(c) Both the Dehn fillings $Y(\cdot, \mu_B, \cdot)$, and $Y(\cdot, \cdot, \mu_G)$ have fundamental group $\mathbb{Z}^2$. (This gives the Hopf link condition.)

(d) The curve ν_B is 0 in $H_1(Y(\mu_R, \cdot, \mu_G); \mathbb{Z})$ and ν_G is 0 in $H_1(Y(\mu_R, \mu_B, \cdot); \mathbb{Z})$. (This gives that $b = g = 0$; note that $r \in \mathbb{Z}$ is automatic since (μ_R, ν_R) is a framing for the red cusp.)

You can check that these four conditions are equivalent to the cores of the fillings in $Y(\mu_R, \mu_B, \mu_G)$ being the needed super-special RBG link. Our method for searching for such Y was:

(1) We looked at simple curves in the dual 1-skeleton of the triangulation of E_K that were not null-homotopic. For each such curve, we drilled it out to get a 2-cusped manifold X. The cusp of X coming from E_K will be called blue and the new one green. The blue cusp inherits a framing (μ_B, ν_B) from the framing (μ_K, λ_K) for E_K.

(2) Check if the filling $X(\nu_B, \cdot)$ is homeomorphic to $E_{K'}$. If so, give the green cusp of X the framing (μ_G, ν_G) inherited from the framing $(\mu_{K'}, \lambda_{K'})$ of $E_{K'}$ and proceed to the next step. Otherwise, go back to Step 1. If we succeed, X will be $Y(\mu_R, \cdot, \cdot)$.

(3) For X passing Step 2, we look through simple curves in the dual 1-skeleton to its triangulation. For each, we drilled it out to get a 3-cusped manifold Y. On the new "red" cusp, we let ν_R (sic) be the curve so that filling along it gives X.

    Provided that requirement (c) holds, we consider $Y(\cdot, \mu_B, \mu_G)$ and test the finitely-many hyperbolically plausible slopes μ_R to see if any $Y(\mu_R, \mu_B, \mu_G)$ is S^3. If so, we check whether (d) is satisfied.

(4) If we succeeded in Step 3, apply [DOR] to get a link L in S^3 whose complement in Y where (μ_R, μ_B, μ_G) are the usual meridians for L. This is the needed super-special RBG link.

Note that this method is not symmetric in K and K', and sometimes it was necessary to switch their roles to find L.

5.12 Remark. As described in Section 5.1, our 0-friend pairs each come from a pair of simple closed geodesics in the hyperbolic structure of their common 0-filling Z_K. At least assuming these geodesics are disjoint, one could search for the red knot among other closed geodesics in Z_K. Our experience was that this approach is not very effective, and it was often necessary to consider the above larger pool of candidates for the red knot. Moreover, unlike Section 5.4, we know of no heuristic reason why the method of Section 5.11 should typically suffice to find such an L, even though it did in all 36 of the examples at hand.

5.13 Remaining mystery 0-friends. We now turn to an expanded version of Theorem 1.12 from the intro.

5.14 Theorem. *If* $18nh_{00000601}$ *is not smoothly slice, or if any of* $\{16n68278$, $17nh_{0010647}$, $18nh_{00098198}\}$ *are smoothly slice, then there is an exotic smooth 4-sphere. All four knots are topologically slice. The knot* $18nh_{00000601}$ *is moreover smoothly slice in a homotopy 4-ball.*

Proof. If the hypothesis holds, then from Table 11 we have a pair of 0-friends where one is smoothly slice and the other is not. This gives an exotic smooth 4-sphere as discussed in e.g. [MP, §1]. To sketch the idea, suppose K and K' are 0-friends where K bounds a smooth 2-disk D in D^4 and K' is not smoothly slice. Let $A = D^4 \setminus \nu(D)$ and B be $S^3 \times I$ with a 0-framed 4-dimensional 2-handle attached to $K' \times \{1\}$. Then ∂A and one component of ∂B are both $Z_K \cong Z_{K'}$; gluing these together gives a manifold W with $\partial W = S^3$. Now W is a homotopy 4-ball and $K' \subset \partial W$ bounds a disk in W, basically the core of the 2-handle used to build B. As K' is not smoothly slice, W is an exotic smooth 4-ball, as needed.

The three knots $\{16n68278, 17nh_{0010647}, 18nh_{00098198}\}$ are topologically slice since they all have $\Delta_K = 1$ [FQ, §11.7]. For $K' = 18nh_{00000601}$, we know it is a 0-friend of a smoothly slice knot K. By the preceding paragraph, K' is smoothly slice in a homotopy 4-ball W. Since every homotopy 4-ball is *topologically* standard [FQ], it follows that K' topologically slice in D^4, as needed. □

knot	DT code of RBG link	r	K' notes
$18nh_{00000601}$	ycjkdnhQyUtMsaVweFIRCXOBgLJDkP	1	ribbon
$16n68278$	tcbjhTDJBgQmsOLhPcekIRFnA	1	$\tilde{\mathbf{s}}_c \neq 0$
$17nh_{0010647}$	scdifSmJkOBcNQPFrdAHEIgl	-1	$\tilde{\mathbf{s}}_c \neq 0$
$17nh_{0010647}$	xcdhlXqLTCvumPWsoAikRHbnEDfJg	-1	$\tilde{\mathbf{s}}_c \neq 0$
$18nh_{00098198}$	rcbgiHMRfPlnJAIOdBcqEKg	-1	$\tilde{\mathbf{s}}_c \neq 0$

Table 11. Five more 0-friends with super-special RBG links, where the original knot in $\mathcal{PS}_{19}$ remains a mystery. The two K' for $17nh_{0010647}$ are distinct.

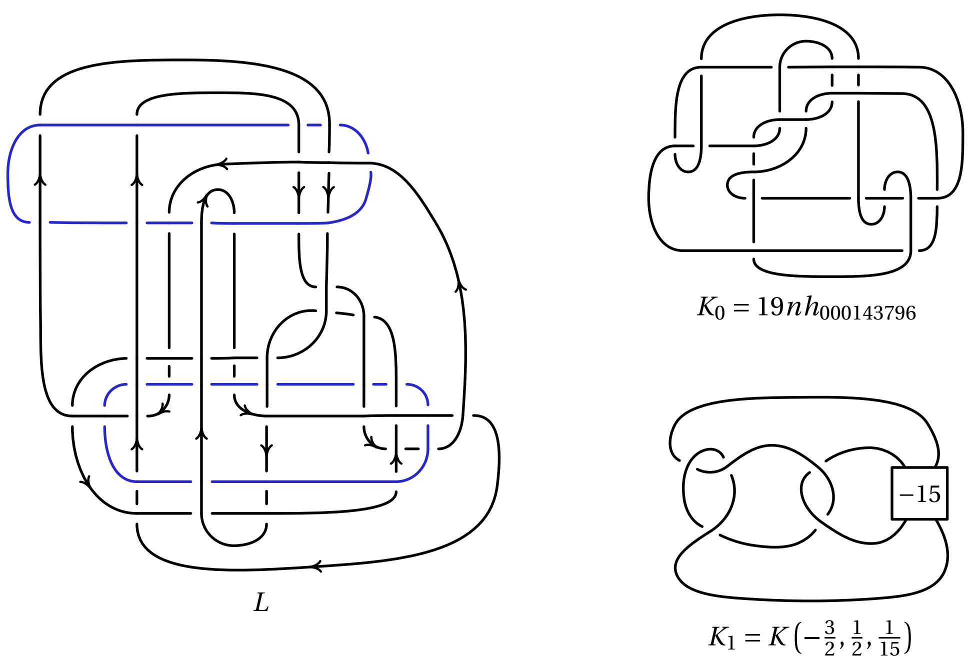


Figure 20. The links from Theorem 6.3. The standard projection of K_0 is shown top right. The knot K_1 is the Montesinos $K(-3/2, 1/2, 1/15)$, so the labeled box is filled in with 15 negative half-twists.